2. Social Network Structure, Measures & Visualization

Contents

Basics of social network structure

- Social network structure refers to the **patterns of relationships** and **connections among a group of people or organizations.**
- Studying social network structure helps us understand how information, resources, and support flow within a group, and how this flow impacts the behavior and outcomes of individuals and organizations.
- Social network analysis is a research methodology that uses **mathematical and statistical techniques to analyze and visualize social network data.**
- By studying social network structure, we can **gain insights into the structure and dynamics of social systems,** and how these systems influence the behavior and outcomes of individuals and organizations within them.

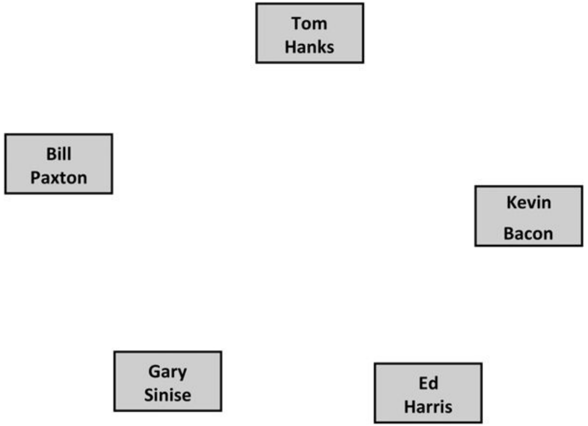
Node in Social Media Network

- A node refers to an individual, organization, or group that is represented by a point in the network.

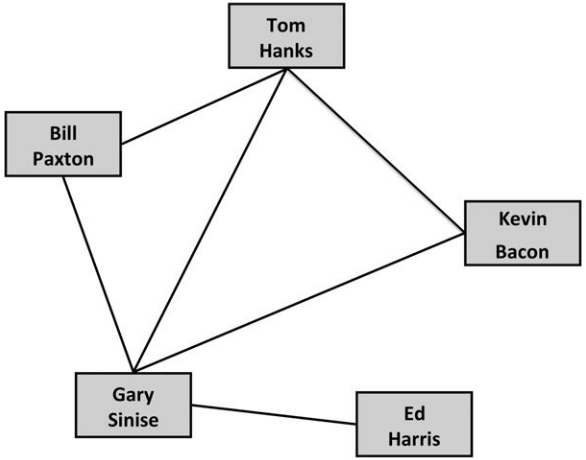
  A node is connected to other nodes by lines or edges, which represent relationships or connections between the nodes.

  These connections can be based on various factors, such as friendships, familial relationships, shared interests, or professional connections.

Node in Social Media Network

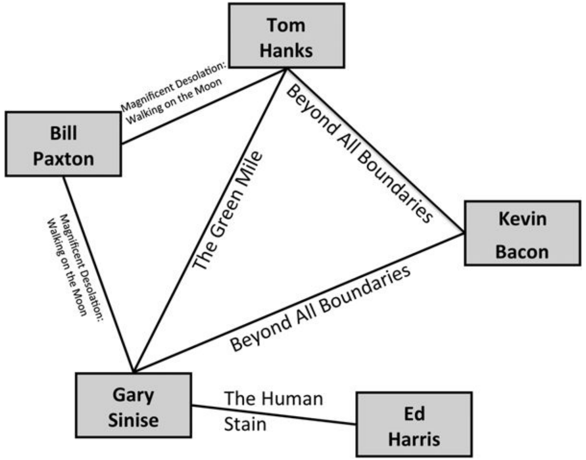


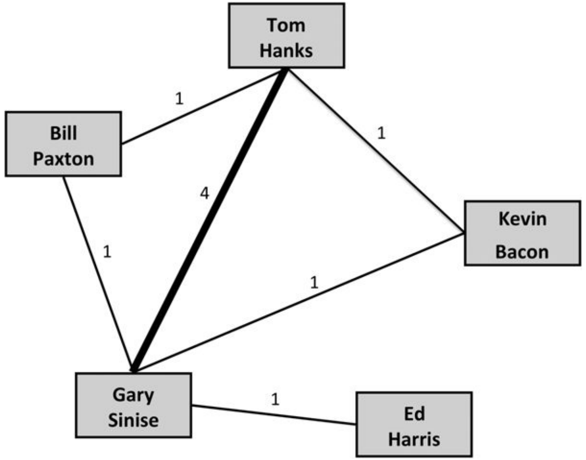
The five co-stars of *Apollo 13*. Each is represented as a node in the network.



The edges connect actors who were in movies together.

Undirected Graph

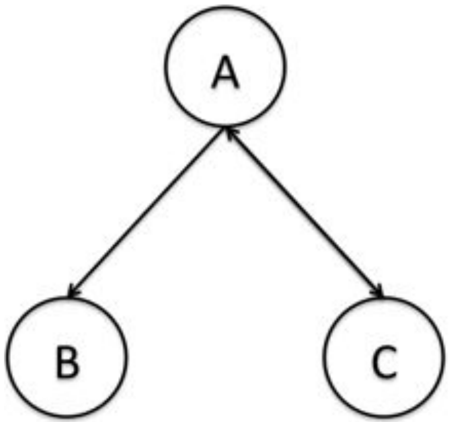


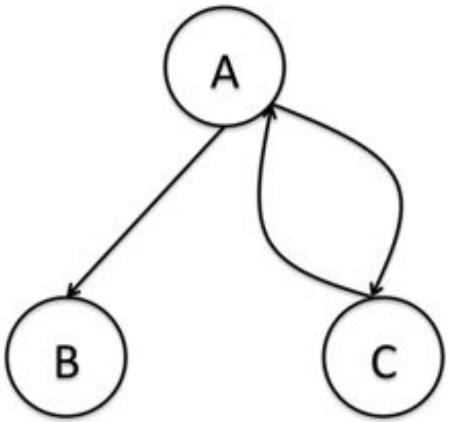


Edges indicate at least one movie that the actors have been in together, not including *Apollo 13*.

A weighted graph where weights are indicated both as numbers and by the thickness of the edge. In this graph, weight indicates how many movies the actors have been in together.

Directed Graphs



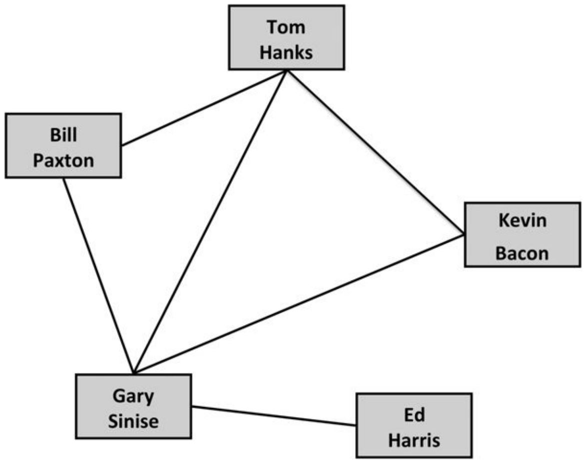


Representing Networks : Text Based Visualizations

- **Adjacency Lists**

An *adjacency list*, also called an *edge list*. Each edge in the network is indicated by list- ing the pair of nodes that are connected. For example, the adjacency list for the *Apollo 13* network is as follows:

- Tom Hanks, Bill Paxton
- Tom Hanks, Gary Sinise
- Tom Hanks, Kevin Bacon
- Bill Paxton, Gary Sinise
- Gary Sinise, Kevin Bacon
- Gary Sinise, Ed Harris

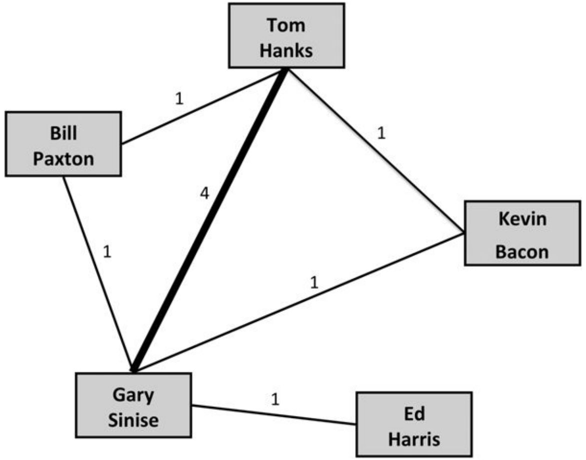


Representing Networks : Text Based Visualizations

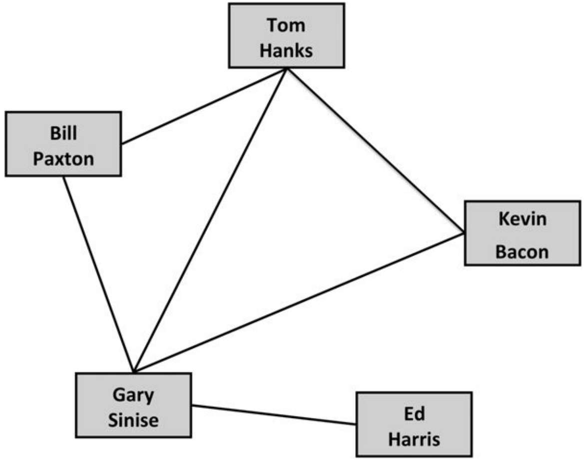
- **Adjacency Lists**

  Adjacency lists can also include additional information about the edges

  - Tom Hanks, Bill Paxton, 1
  - Tom Hanks, Gary Sinise, 4
  - Tom Hanks, Kevin Bacon, 1
  - Bill Paxton, Gary Sinise, 1
  - Gary Sinise, Kevin Bacon, 1
  - Gary Sinise, Ed Harris, 1



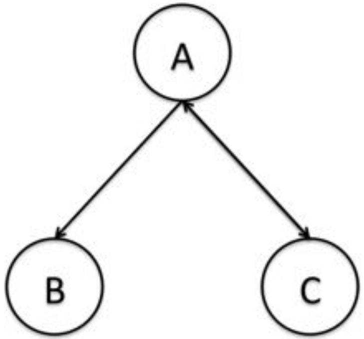
Adjacency Matrix



The Adjacency Matrix for the Apollo 13 Network

	Tom Hanks	Bill Paxton	Gary Sinise	Kevin Bacon	Ed Harris
Tom Hanks	0	1	1	1	0
Bill Paxton	1	0	1	0	0
Gary Sinise	1	1	0	1	1
Kevin Bacon	1	0	0	0	0
Ed Harris	0	0	1	0	0

Adjacency Matrix for directed graph

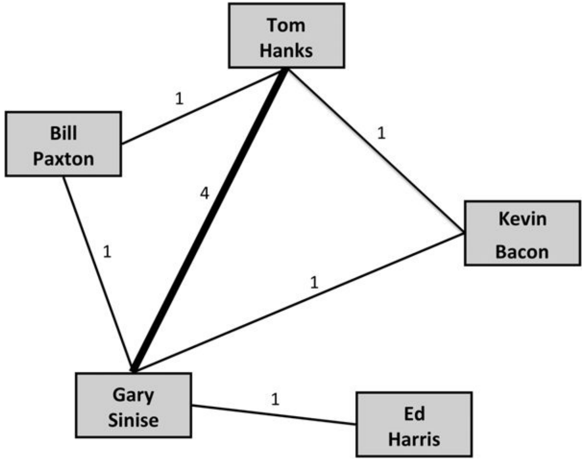


Fig(1)

A Small Adjacency Matrix for a Directed Network

	A	B	C
A	0	1	1
B	0	0	0
C	1	0	0

Adjacency Matrix for Undirected Weighted Graph



Fig(1)

The Adjacency Matrix for the Apollo 13 Network with Edge Weights

	Tom Hanks	Bill Paxton	Gary Sinise	Kevin Bacon	Ed Harris
Tom Hanks	0	1	**4**	1	0
Bill Paxton	1	0	1	0	0
Gary Sinise	**4**	1	0	1	1
Kevin Bacon	1	0	0	0	0
Ed Harris	0	0	1	0	0

XML and standard format

```
<Person>
    <name>Tom Hanks</name>
    <connection>Bill Paxton</connection>
    <connection>Gary Sinise</connection>
    <connection>Kevin Bacon</connection>
</Person>
```

Basic Network Structure and Properties

Subnetworks

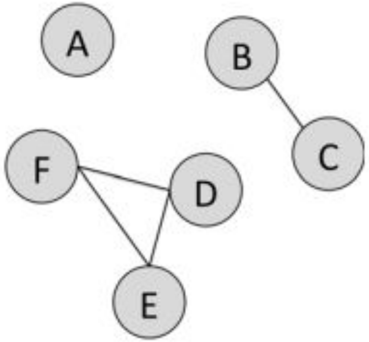


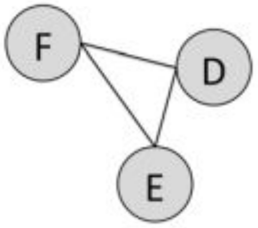
FIGURE 2.6 A social network with a singleton, dyad, and triad

Subnetwork Types

- Singleton
- Dyad
- Triad

Cliques

- All nodes in a group are connected to one another. When this happens, it is called a clique.



Egocentric networks

As we are going one step away from D in the network, this is called a degree-1 egocentric network. It only shows us the nodes D is connected to.

If we want to see only D’s neighbors and their connections, it is called a 1.5-degree egocentric network

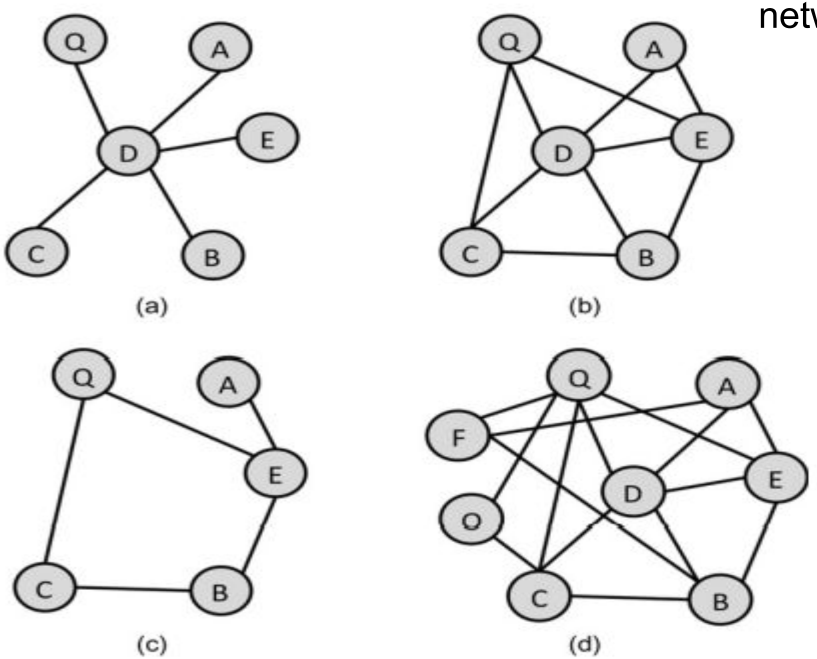


FIGURE 2.8

(a) The 1-degree egocentric network of D, (b) the 1.5-degree egocentric network of D, (c) the 1.5 egocentric network of D with D excluded, and (d) the 2-degree egocentric network of D.

Paths and Connectedness

- A path is a series of nodes that can be traversed following edges between them.
- To determine the length of a path, we count the number of edges in it. The path from M to C has a length of 4 (MP, PF, FO, and OC)

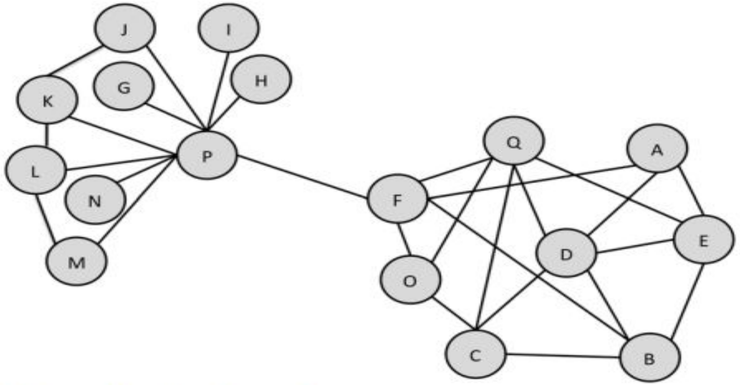


FIGURE 2.7 A sample undirected network

Shortest Path

- There are two shortest paths from Node F to Node E: FAE and FBE. Shortest paths will be an important measure we consider in network analysis and are sometimes called geodesic distances

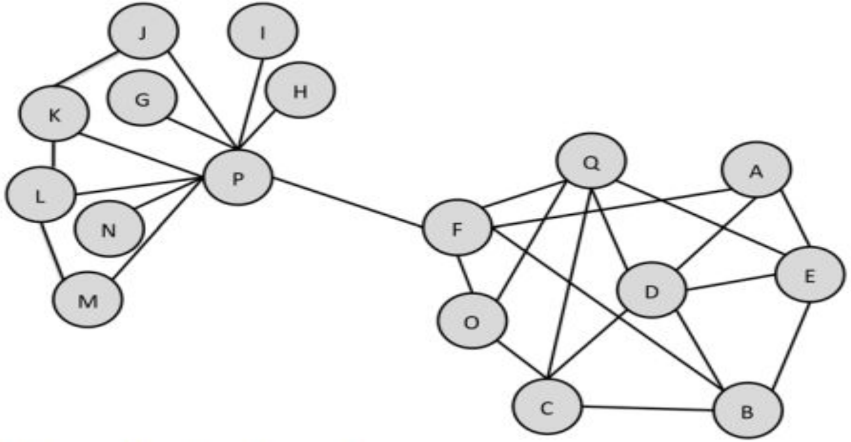


FIGURE 2.7 A sample undirected network

Connectedness

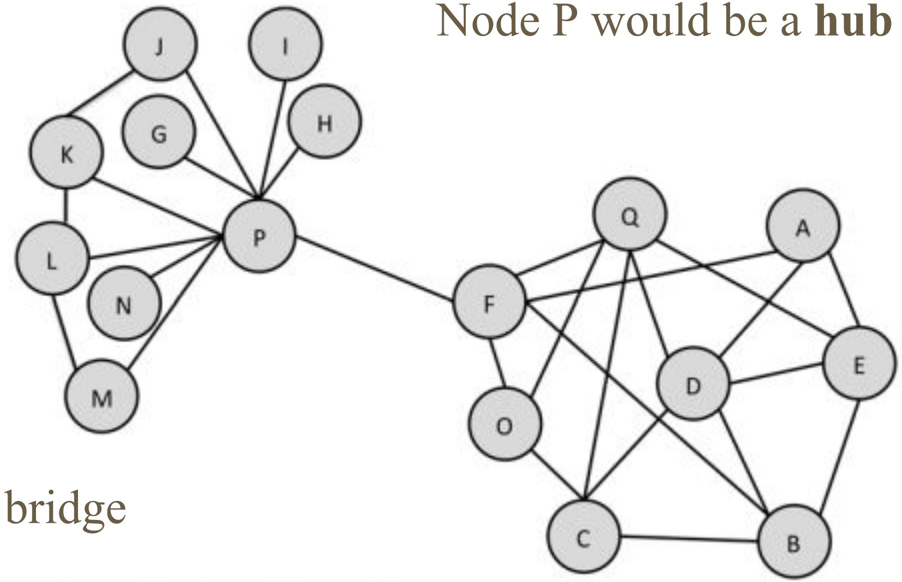
- Paths are used to determine a graph property called connectedness. Two nodes in a graph are called connected if there is a path between them in the network.
- An entire graph is called connected if all pairs of nodes are connected.
- **Strongly connected graphs** - every vertex is reachable from every other vertex.
- **Weakly connected** - If a path cannot be found between all pairs of nodes using the direction of the edges, but paths can be found if the directed edges are treated as undirected, then the graph is called weakly connected.

Bridges and hubs

- **Connected Components** -If a graph is not connected, it may have subgraphs that are connected. These are called connected components. The fig shows three-node connected component, a two-node connected component, and a singleton
- **A bridge is** an edge that connects two otherwise separate groups of nodes in the network. Formally, a bridge is an edge that, if removed, will increase the number of connected components in a graph.
- **Hub** - The term is used to refer to the most connected nodes in the network. E.g. Node P would be a hub because it has many connections to other nodes.

Bridges and hubs

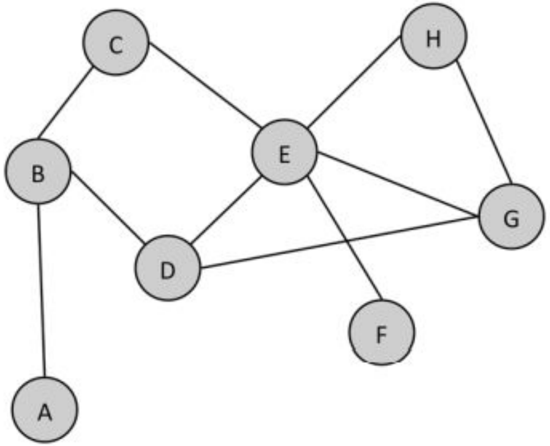
Node P would be a **hub**



PF is a bridge

FIGURE 2.7 A sample undirected network

Exercise



1. Answer the following questions about this graph.

a. How many nodes are in the network?

b. How many edges are in the network?

c. Is this graph directed or undirected

d. Create an adjacency list for this graph.

e. Create an adjacency matrix for this graph

f. What is the length of the shortest path from node A to node F?

g. What is the largest clique in this network? How many cliques of that size are there?

h. How many connected components are there in this network?

i. Draw the 1.5 ego network for node E (without including node E in the graph). How many singletons are in the ego network?

j. Are there any hubs in the network? If so, which node(s) and why is it a hub?